AI in Healthcare Hackathon 2026 — Participant Guidelines

Host: Central Asian University
Start: March 26, 2026 — 08:00 (Tashkent Time, UTC+5)
Final Submission Deadline: March 27, 2026 — 20:00 (Tashkent Time)
Top 10 Teams Announcement: March 27, 2026 — 22:00
Top 10 Presentations: March 28, 2026 — 09:30 – 14:30

Presentation Language: English
Note: One member of a team may present on behalf of the entire team.

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1. Objective

The **AI in Healthcare Hackathon 2026** challenges participants to develop artificial intelligence systems capable of analyzing **medical radiograph images**.

Participants will:

- Develop AI models to solve **two machine learning problems**
- Work with real medical imaging datasets
- Apply deep learning and image processing techniques where appropriate
- Generate outputs using trained models
- Demonstrate their methodology and system design

The competition evaluates:

- Technical capability
- Model design
- Implementation strategy
- Practical AI system development

2. Dataset Access

Participants must download the dataset using the following links.

Primary dataset link (Link will be provided on the day of starting of hackathon)

Important

After downloading the dataset:

- Extract all files
- Ensure the dataset downloaded successfully
- Verify that no files are missing before beginning work

The dataset structure and required output formats will be explained within the dataset package.

3. Dataset Description

The dataset contains **medical images** intended for AI-based analysis.

Participants will solve **two tasks**:

Problem A

Participants must develop a model that analyzes images and produces **structured outputs for each image** according to the format specified in the dataset template.

Problem B

Participants must develop a model that **generates output files corresponding to each input image**.

Participants should carefully examine the dataset and template files to understand the expected outputs.

4. Tasks

Participants must complete **both Problem A and Problem B**.

Problem A

Participants must train a model using the training data provided for **Problem A** and generate outputs for the corresponding test images.

Outputs must follow the requirement that will be provided at the start of the hackathon.

Problem B

Participants must train a model using the training data provided for **Problem B**.

Outputs must follow the requirement that will be provided at the start of the hackathon.

5. Submission Instructions

Participants must upload all required files through the **Google Form**.

Submission form:

Will be provided at the start of the hackathon

A QR code for the submission form will also be provided.

Participants will upload **six files separately**.

6. Required Submission Files

Result File for Problem A

You will be required to follow the requirement provided at the start of the hackathon

Output Files for Problem B

You will be required to follow the requirement provided at the start of the hackathon

Inference Script for Problem A

Upload the inference script used to generate the outputs for Problem A.

You will be required to follow the requirement provided at the start of the hackathon

The script must:

- Load the trained model
- Accept a directory containing test images
- Generate the required result file

The script must perform **inference only** and must **not retrain the model**.

Inference Script for Problem B

Upload the inference script used to generate outputs for Problem B.

You will be required to follow the requirement provided at the start of the hackathon

The script must:

- Load the trained model
- Accept a directory containing test images
- Generate the required output files

The script must perform **inference only**.

Model for Problem A

Upload the trained model used for Problem A.

You will be required to follow the requirement provided at the start of the hackathon

The model must correspond to the submitted **Python** script.

Model for Problem B

Upload the trained model used for Problem B.

You will be required to follow the requirement provided at the start of the hackathon

The model must correspond to the submitted **Python** script.

Important Notes

- All filenames must include the **exact Team Name used during registration**
- Scripts are required **only for verification purposes**

- Scripts must **not retrain models**

7. Data Usage Rules

Participants may use:

- Pretrained models
- Transfer learning
- Public open-source libraries
- Public datasets for **pretraining or fine-tuning**

The following are strictly prohibited:

- Using the provided **test data for training**
- Manually modifying outputs for the test set
- Sharing predictions, models, or solutions with other teams

Organizers may request explanations or code to verify results.

Violation may lead to **disqualification**.

8. Scoring Overview (Total: 100 points)

Phase 1 — Model Performance (70 points)

Model performance will be evaluated using **standard evaluation metrics appropriate for each task**.

- **Problem A performance:** 30 points
- **Problem B performance:** 40 points

Exact evaluation metrics will be applied automatically during scoring.

The **Top 10 teams** based on Phase 1 scores will advance to the presentation stage.

Phase 2 — Top 10 Presentation Evaluation (30 points)

Shortlisted teams will be evaluated based on:

Category	Points
User Interface Design	10
Error Handling	5
Innovation & Creativity	5
Presentation & Clarity	10

Final rankings will be determined using the **combined score from Phase 1 and Phase 2**.

The **top three teams will receive prizes**.

9. Presentation Guidelines (Top 10 Teams)

- Presentations will be conducted **in English**
- **One team member may present on behalf of the team**

Each team must clearly explain their approach.

The presentation must include:

AI Workflow

- Overall system pipeline
- Data preprocessing steps
- Training and inference procedures

Model Design

- Models or architectures used
- Explanation of why those models were selected
- Training strategy (transfer learning, pretrained models, etc.)

Challenges and Solutions

- Key challenges encountered during development
- Methods used to address those challenges

Results

- Observations from experiments
- Performance insights
- Model behavior analysis

Interface Demonstration

Teams must demonstrate an interface showing how their AI system works.

The interface should allow visualization of:

- Image input
- Model outputs
- Output visualization

Allowed formats

- Web application
- Desktop application
- Dashboard
- Interactive prototype

Important Requirement

The interface must clearly display the statement:

“For research and demonstration purposes only. Not for clinical use.”

10. Judging and Quality-Based Evaluation

In addition to automated performance evaluation, the judging panel will assess the **overall quality, robustness, and understanding of the proposed solution** during the presentation stage.

During the Top 10 presentations, judges may evaluate:

Technical Understanding

- The team’s understanding of the methods used
- Ability to explain the model design and workflow
- Ability to answer technical questions about implementation

Solution Quality

- Soundness of the approach
- Proper handling of data and outputs

- Reliability and stability of the system

System Demonstration

- Clarity and usability of the interface
- Correct functioning of the AI system during demonstration
- Ability to reproduce outputs using the trained model

Innovation and Engineering Quality

- Creative use of AI techniques
- Efficiency of the proposed pipeline
- Practical design considerations

Judges may ask **technical questions related to the model, data processing, and implementation** to verify the authenticity and understanding of the solution.

Teams are expected to demonstrate **clear knowledge of their approach and system design**.

Failure to adequately explain the system may affect the final evaluation.

11. Academic Integrity and Authenticity of Work

All submitted solutions must represent the **genuine work and understanding of the participating team**.

Teams are expected to:

- Clearly understand the methods, models, and code used in their solution
- Be able to explain their implementation and design decisions during the presentation and Q&A session
- Demonstrate how their system processes input data and generates outputs

During the evaluation stage, judges may ask **technical questions about the model, data processing pipeline, implementation details, and system design** to verify the authenticity of the work.

Teams must be able to **clearly explain the methodology and implementation used in their system**.

The following activities may lead to disqualification:

- Submitting work that team members cannot explain or demonstrate

- Using solutions obtained from other teams or external participants
- Copying code or models without proper understanding
- Sharing models, outputs, or solutions with other teams
- Manipulating outputs or evaluation results

If the judging panel determines that a team **cannot adequately explain their solution or demonstrate understanding of their implementation**, the team may be **disqualified from the competition**.

12. Important Notes

- Internet access is allowed
- Pretrained models are allowed
- Participants can use TensorFlow or PyTorch
- Focus on **model performance, methodology, and clarity**
- Judges' decisions are final

13. Deadlines & Announcements

Final submission deadline:

March 27, 2026 — 20:00 (Tashkent Time)

Top 10 teams announcement:

March 27, 2026 — 22:00

Top 10 presentations:

March 28, 2026 — 09:30 – 14:30